SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA14 **COURSE NAME: Compiler Design**

COURSE OUTCOME COVERED

CO4: *Examine intermediate code generation techniques to enhance code optimization (BL4)*

Assessment Tool Weightage: 40%

QUESTIONS: 4 **Total Marks:40**

Annexure A

CLASS TEST

Code of Conduct:

I _B.MOKSHA SREE (192511160)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: B.Moksha

Q. No	Understanding of Concepts (25)	Explanation (25)	Applicatio n (20)	Organizatio n(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	**/ 30**	**/ 20**	**/ 15**	**/ 10**	**/ 100**

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconception s	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

Name:- B.Moksha sree

reg NO:- 192511160 "Cou-Ati"

Course:- Compailer design

Course code:- CSA1402 Class test

(Q1) if ((a< b) and ((c<d) or (a>d))) then

z = x+y*z

else

z=z+1

38/40

(A) if ((a< b) and ((c<d) or (a> d))).

z = x+y*z

else

z = z+1

three adress code for the statement

L1:- if a<b goto L2

goto L5

L2:- if c<d goto L4

goto L3

L3:- if a>d goto L4

goto L5

L4:- t1=y*z

t2=x+t1

z=t2

goto L6

L5:- z=z+1

L5:-

t3=z+1

z=t3

2) Translate (a< b) and (c<d) or (e<f) into three adress statements using back Patching.

$[(a<b) \text{ and } (c<d)] \text{ or } (e<f)$

1) if (a< b) goto 3
2) goto 5
3) if (c<d) goto 7
4) goto 5
5) if (e<f) goto 7
6) goto 8
7) True
8) False.

(3) $A = (-A+B) * ((-A+B) * C)$

$t_1 := -A$

$t_2 := t_1 + B$

$t_3 := -A$

$t_4 = t_3 + B$

$t_5 = t_4 * C$

$t_6 = t_2 * t_5$

$A = t_6$

"three adress code"

for the Expression

$A = (-A+B) * ((-A+B) * C)$

OP	arg 1	arg2	result
Unry	A	-	t_1
+	t_1	B	t_2
			t_3
Unry	A	-	t_4
+	t_3	B	
*	t_4	C	t_5
*	t_2	t_5	t_6
=	A6	-	A

(Q4) Construct three adress code using the following Production for Boolean Expression. (a>b) or (c<d) or (e<

$B_1 \rightarrow B_1 || B_2$
$B \rightarrow B_1 \&\& B_2$
$B \rightarrow ! B_1$
$B \rightarrow E_1$ relop E_2
$B \rightarrow$ true
$B \rightarrow$ false

L1:- If (a>b) goto LTRUE
go to ~~ELAND~~ L2

L2:- If (c<d) goto LTRUE
go to L3

L3:- If (e<f) goto LTRUE
goto LFalse.

LTRUE:-
//Boolean Expression is true
goto LEND

LFALSE:-
//Boolean expression is false

LEND: